

AI OPS Assignment 2

Pulkit Jindal da24b047

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1 Question 1

The Image size for the single stage build is about 1.51 GB while for the multi stage build it is around 1.43 GB. For both the builds the considered base image was Python non slim image. The reduction in the image size was around 5%. This is a noticeable drop in the memory required for the image and this memory required has majorly decreased because in the first stage we are taking the entire python notebook as well as excess dependencies file and then building the given model while in the multi stage build the second stage only contains the model and its required dependencies. Due to this there are some files which are not present in the second stage as they are unnecessary for the final model and thus save on memory.

2 Question 2

Docker Compose allows multiple independent containers to interact with each other which allows containers to have more capabilities. For example in this question the docker file was able to make use of the extra redis container which served as a database for the model to store its predictions. This is a good example of having multiple containers work together. On the other hand Kubernetes is much more large scale and responsible for allocating work to a large number of containers across multiple nodes which are not necessarily interacting with each other. Docker Compose also works only on one "node" or machine while Kubernetes runs across many nodes.

3 Question 3

No our Parallelism choice for this remains unchanged because we have 8 total tasks and having a parallelism of 8 will allow us to make full use of our nodes. Earlier we had 2 nodes on 2 cpus each and that would have been slower as we have less nodes running and having 3 nodes will be faster but the parallelism will not need to be changed it will remain 8.

4 Question 4

- The Replica Set Controller is responsible for ensuring that the pods are all running and that if they are deleted that they are restored. It decides if a replacement is needed if the current number of pods running differs from the number of desired pods to be running.
- Had the update on the container been done in the case of Job Completion there would be no rolling update applied to because it differs from Deployment in not being constantly live and running.
- Deployment Self Healing and Rolling Update differs from the Job completion because the deployed pods are meant to run indefinitely until they are stopped and thus if the pods are down for some reason they must also be restored back to their original state and similarly if there is an update applied to the pods it must reflect in the live state of the pods. Meanwhile on the other hand the Job has a finish state which is when the pods finish all the tasks and complete them and thus they a job is very different from deployment.